

QML: The Next Big Thing

By

Shakthi Ganesh

QML Developer, KwantumG Research Labs Pvt Ltd

Mathematics for QML

- Classical ML is hitting scale limit
- Rising Compute and energy cost
- Some problems don't scale well classically
- Quantum Computers offer a different approach
- Why this matters for ML specifically?

Classic ML vs Quantum ML

- Data Representation: bits vs qubits
- Core computation: Matrix operations vs unitary transformations
- How information is extracted
- Hardware maturity
- Current maturity and readiness
- Where each approach fits best

Quantum Advantage in ML

- Quantum speedup means solving a problem in fewer computational steps than any known classical method
- Quantum kernel methods can access feature spaces that are classically hard to compute
- Sampling problems are a strong candidate for quantum advantage
- Optimization landscapes may be explored more efficiently with quantum methods
- Advantage is problem-dependent, not universal

Current Industry Trends and Adoption

- Major tech companies are competing to build practical quantum hardware and cloud access
- Cloud quantum platforms have made experimentation accessible without owning hardware
- Investment in quantum computing and QML startups has grown significantly
- Adoption today is concentrated in pilot programs, not full production deployment
- Industry consortiums and standards bodies are starting to shape the field

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By

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Founder/CEO, KwantumG Research Labs Pvt Ltd

Mathematics for QML

- Qubit states are represented as vectors in a complex vector space
- Quantum gates are represented as unitary matrices
- Multiple qubits combine via tensor products
- Measurement is described by projection and probability via the Born Rule
- Parameterized quantum circuits use rotation gates as trainable parameters

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha \begin{bmatrix} \gamma \\ \delta \end{bmatrix} \\ \beta \begin{bmatrix} \gamma \\ \delta \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{bmatrix}$$

Foundations

- Complex Number
- Sets
- Linear Algebra
- Group
- Field
- Space, Vector Space, Hilbert Space
- Eigenvalue and Eigenvector
- Matrix Composition and Decomposition

Complex Numbers

- Qubit is a Two Elements Complex Column Vector

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 0 + 0i \\ 1 + 0i \end{bmatrix}$$



$$\begin{bmatrix} 1.44 \\ 30 \end{bmatrix}$$

- Sub Atomic scale behaviours of physical systems
- We need COMPLEX NUMBER
- Why TWO elements?
 - Spin of Electron
 - Polarity of Photon
- We are choosing Physical systems which are having two basis states
 - Coin toss has two basis states
 - Dice roll has 6 basis states

Complex Numbers Contd...

- Representations
 - Cartesian : $a + ib$
 - Polar : $r<\theta$
 - Exponential : $re^{i\theta}$

Set

- All physical systems are represented using Sets
- Zero ket

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|\psi\rangle = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

- Basis States

$$|0\rangle, |1\rangle,$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Linear Algebra

- Gilbert Strang YouTube, 3b1b YouTube
- Inner Product
 - Output is Scalar
- Outer Product
 - Output is Matrix
- Kronecker Product or Tensor Product
 - Elevated Matrix

$1 \times 0 + 2 \times 0$

$\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 0 \end{bmatrix}$

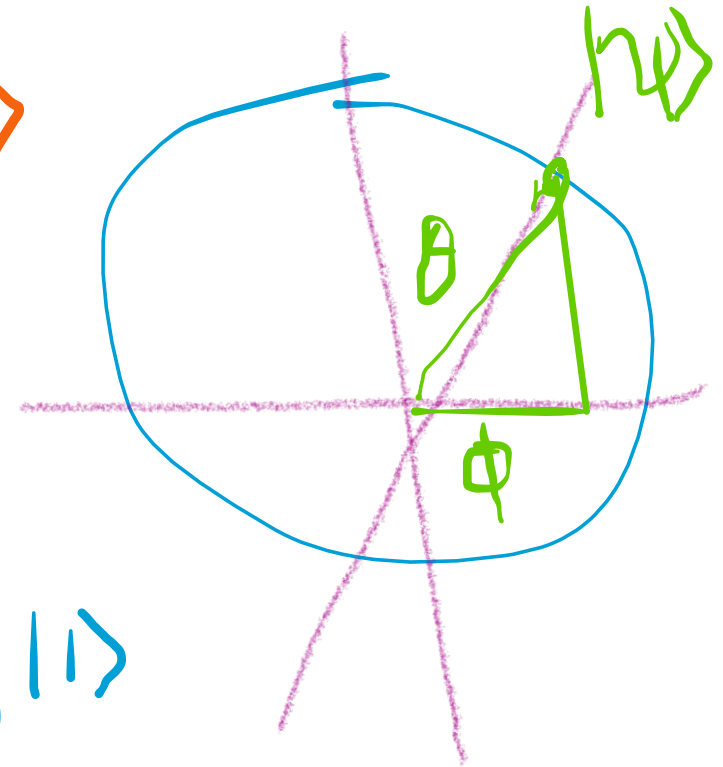
Group, Field, Space, Vector Space, Hilbert Space

- Hilbert space is Complex Vector space where inner product is closed
- Why ? We can use bloch sphere to represent

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha^2 + \beta^2 = 1$$

$$|\psi\rangle = \cos \theta/2 |0\rangle + e^{i\phi} \sin \theta/2 |1\rangle$$



Eigenvalue and Eigenvector

QPE is based on Eigenvalue and Eigenvector

- Characteristics vector and scalar of a matrix

$$A v = \lambda v$$

$$A |a\rangle = \lambda |a\rangle$$

$$U |x\rangle = e^{i\theta} |x\rangle$$

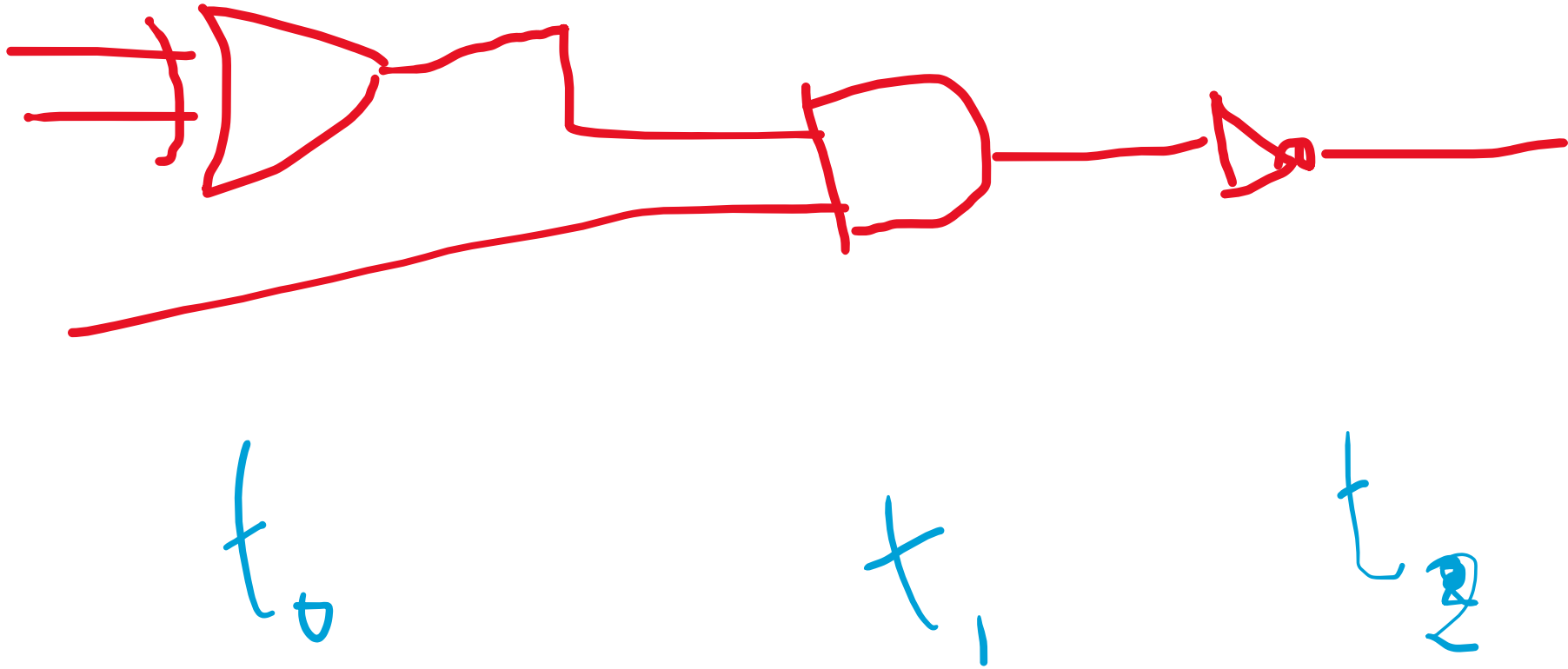
Quantum Gates

- H, X, Y, Z, P, T, Tdg, S, Sdg, U
- CNOT, Swap
- CCNOT, Fredkin
- All quantum gates are Hamiltonian, Hermitian and thus they are Unitary
- Hamiltonian is Energy Operator
- Hermitian is Complex matrix which is equivalent to Transpose of self conjugate

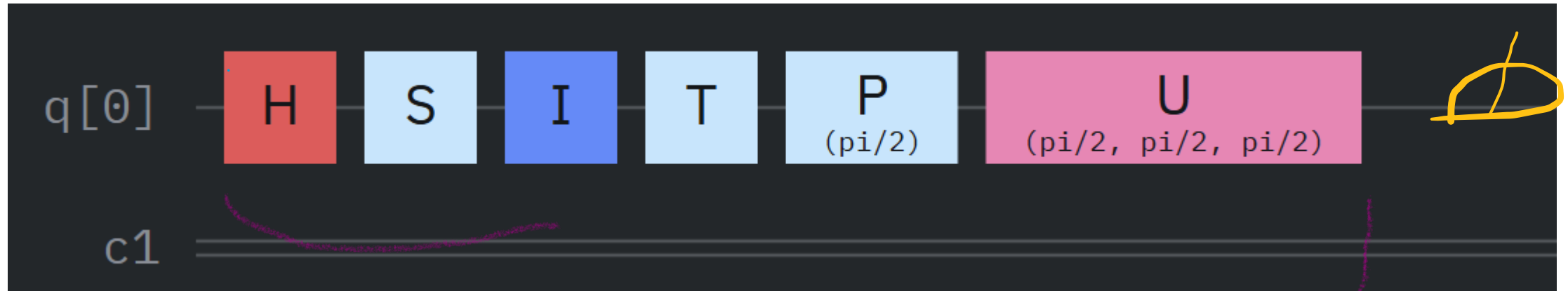
$$A = (A^*)^T$$

Classical circuits need N clock cycle

- XOR + AND + NOT



All quantum gates are unitary matrices, so they are executed as one operation



to

Matrix Composition

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \begin{bmatrix} i \\ 1 \end{bmatrix}$$

Handwritten red annotations below the equation:

A red bracket groups the three matrices. Below this bracket, the expression is written in red:

$$\begin{bmatrix} a \\ c \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} \begin{bmatrix} e \\ f \end{bmatrix} =$$

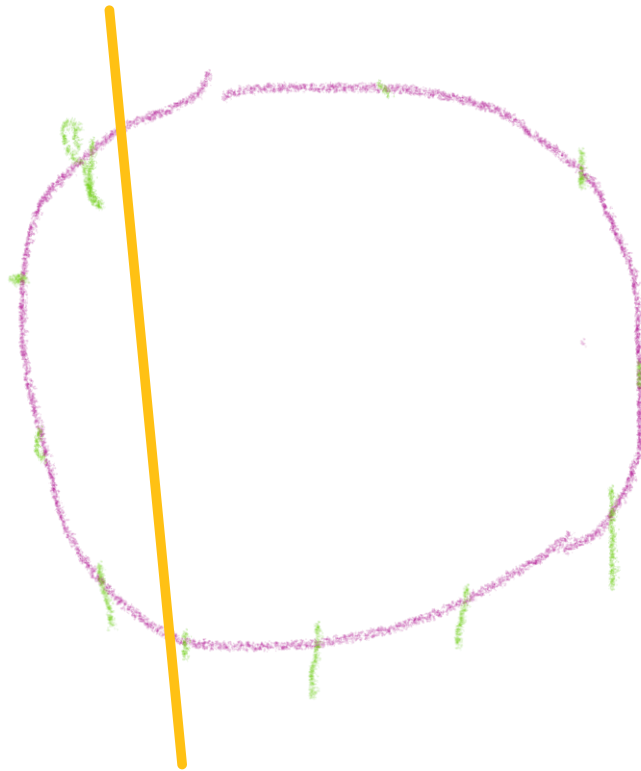
Singular Value Decomposition

- Schmidt Decomposition

- QSVM replaces the classical kernel function with a quantum kernel
- Data is mapped into quantum states using a feature map
- The quantum kernel is computed as the overlap between two encoded states
- The resulting kernel matrix is passed into a standard classical SVM solver
- A simple two-feature example illustrates the workflow end-to-end

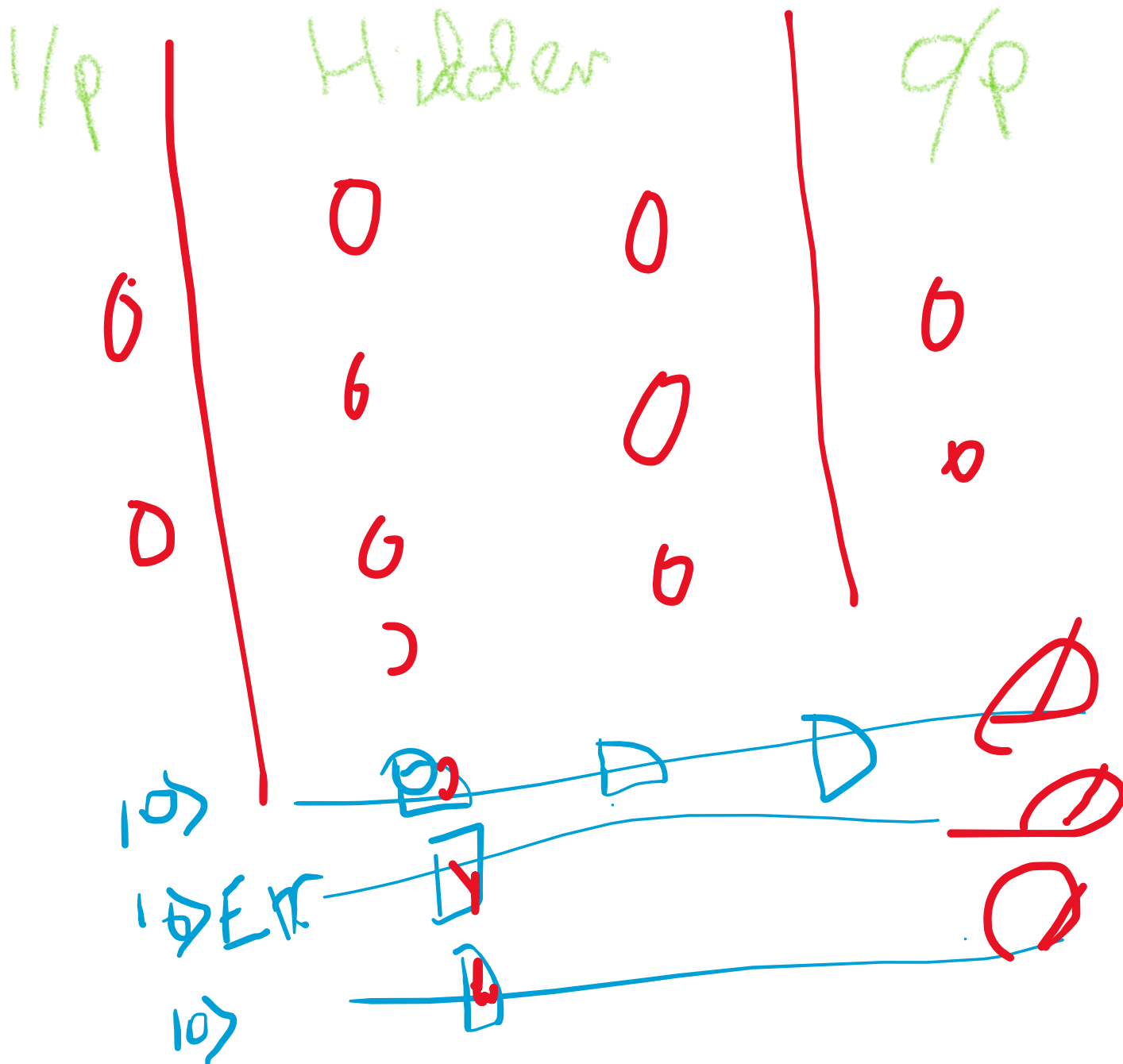
QSVM

- Kernel must be in Hilbert space



- A simple two-feature example illustrates the workflow end-to-end
- Data encoding maps classical inputs into rotation angles
- The variational layer applies trainable rotations and entangling gates
- The circuit's expectation value serves as the model's output
- Training uses the parameter-shift rule to compute gradients

QNN



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Key QML algorithms overview

- VQE estimates the ground-state energy of a system
- QAOA solves combinatorial optimization problems
- QSVM extends support vector machines using quantum kernels
- QNNs generalize neural networks using parameterized quantum circuits
- Quantum kernel estimation underlies several algorithms beyond QSVM

Hybrid Quantum~Classical Models

- Hybrid models split work between quantum and classical processors by design
- This is the practical necessity of the NISQ era
- Classical optimizers drive the training loop
- The quantum-classical loop repeats iteratively
- This design pattern underlies most practical QML today

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Basic Algorithms

- Basic Algorithms
 - Deutsch-Josza
 - Bernstein-Vazirani
 - Simon's
 - Quantum Fourier Transform
 - Quantum Phase Estimation
 - Shor's
 - Grover's
 - Superdense coding
 - Quantum Key Distribution
 - Hidden Subgroup Problem

[Quantum Algorithm Zoo](#)

- Optimization Algorithms
 - QUBO
 - Quadratic Unconstrained Binary Optimization
 - QAOA
 - Quantum Approximate Optimization Algorithm
 - VQE
 - Variational Quantum Eigen Solver
 - HHL
 - Harrow-Hassidim-Lloyd
 - VQLS
 - Variational Quantum Linear Solver

QML Models

- QSVM
- QPCA
- QNN
- QKNN
- QCNN
- QLTSM
- QGAN
- QTL : Quantum Transfer Learning
- QRC : Quantum Reservoir Computing
- QGRNN : Quantum Graph Recurrent NN
- QPINN : Quantum Physics Informed NN

Benchmarking

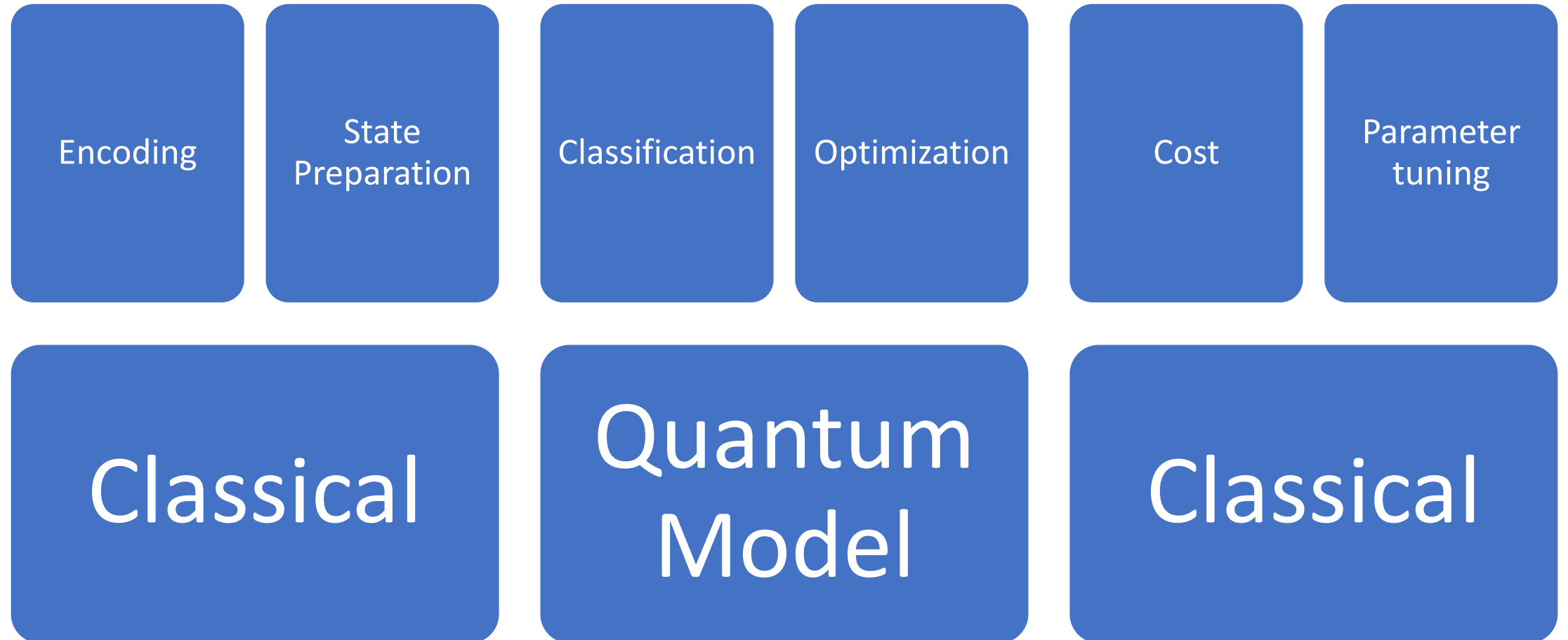
- Classical & Quantum Inspired Model
 - Laptop
 - Cloud GPU
 - On-prem GPU
- Quantum
 - Super Conducting
 - Trapped Ion
 - Neutral Atom



Hybrid Quantum~Classical Models

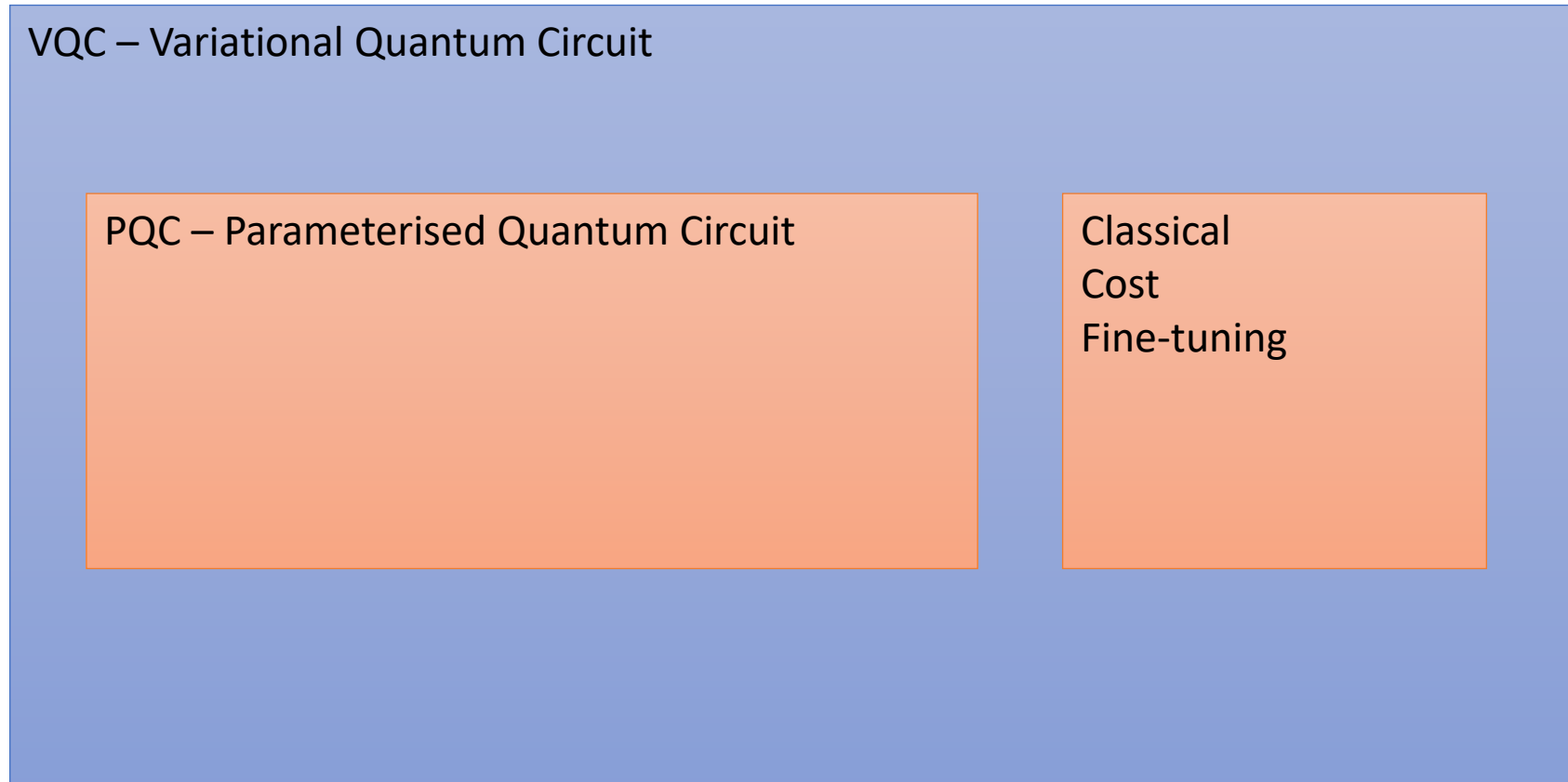
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Hybrid Models



Hybrid Quantum Classical Approach

Variational Quantum Circuit



QML Approaches

- CC – Classical Data Classical Algorithm – existing classical ML
- CQ – Classical Data Quantum Algorithm
- QC – Quantum Data Classical Algorithm
- QQ – Quantum Data Quantum Algorithm
 - Future approach
 - Needs Quantum Sensing

Real-world Use Cases and Applications

- Healthcare & Drug Discovery
- Finance & Risk Analysis
- Logistics & Supply Chain
- Manufacturing & Predictive Maintenance
- Cybersecurity
- Scientific Research

Healthcare & Drug Discovery

- Protein folding
- Faster drug discovery
- Personalized medicine
- Medical image analysis

Finance

- Portfolio optimization
- Fraud detection
- Risk analysis
- Algorithmic trading

Logistics & Manufacturing

- Route optimization
- Warehouse scheduling
- Predictive maintenance
- Production planning

Combinatorial Optimization

- QUBO 1 or 0
- Ising -1 o 1
- 3x3 -> 1 hour
- 4x4 -> 2 hours
- 5x5 -> 4 hours
- 6x6 -> 8 hours

Challenges in QML

- Limited qubits
- Noise & decoherence
- High error rates
- Expensive hardware

Software & Skills Challenges

- Immature ecosystem
- Limited datasets
- Complex debugging
- Need AI + Quantum + Math skills

1.QuLacs

1.C++ and Python

2.Pennylane

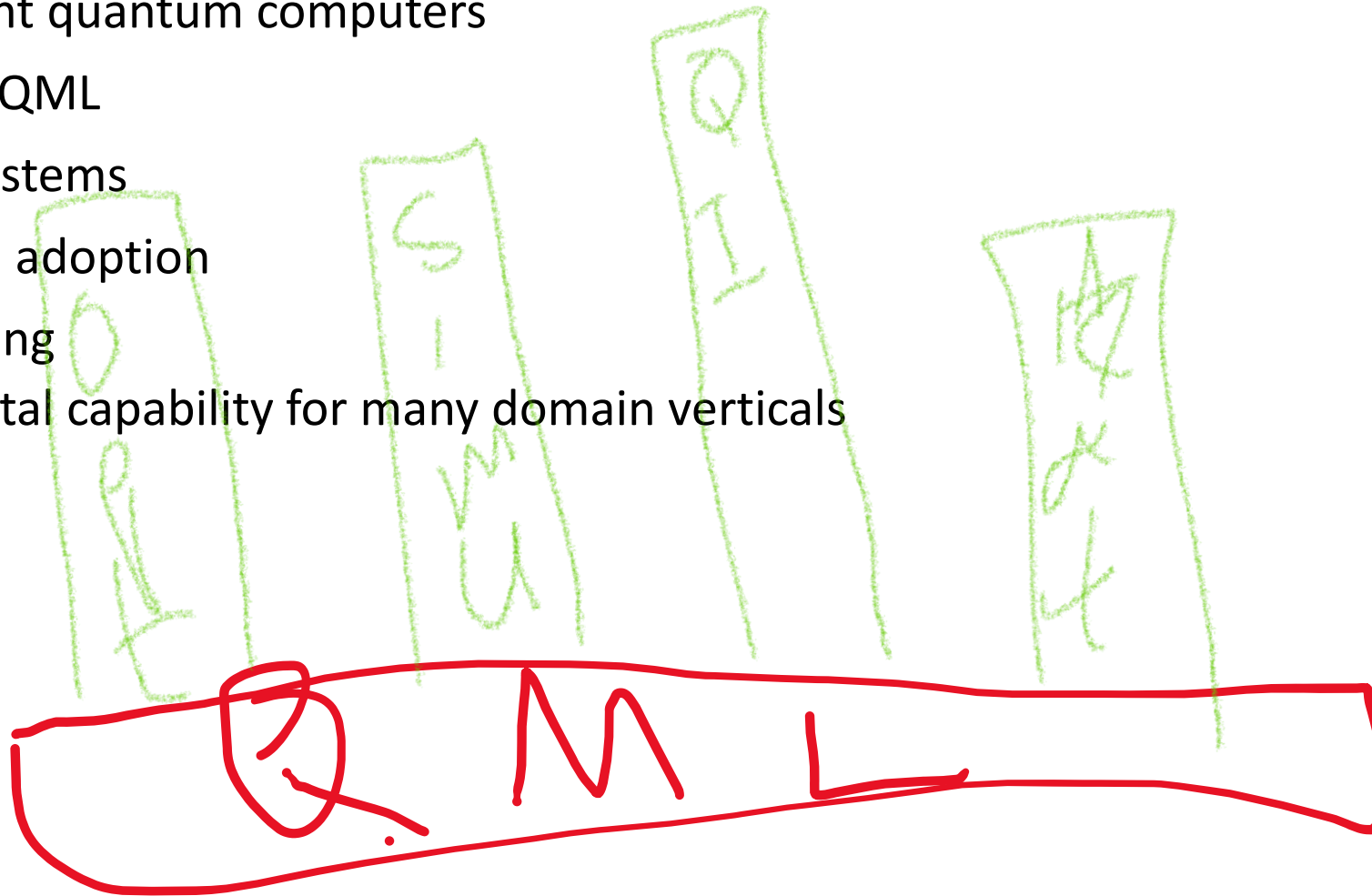
3.Bloqade

1.Julia and Python

4.Qiskit

Future Score

- Fault-tolerant quantum computers
- Large-scale QML
- Hybrid AI systems
- Commercial adoption
- Next Big Thing
 - Horizontal capability for many domain verticals



Career Opportunities

- Quantum ML Engineer
- Quantum Researcher
- Quantum Software Developer
- Quantum AI Consultant

Contact Us

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QML: The Next Big Thing Quantum Reservoir Computing

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Quantum Reservoir Computing

5. sparsely connected NN

- Quantum Reservoir Computing (QRC) uses a fixed quantum system as a high-dimensional computational reservoir
- Only the classical output layer is trained, reducing optimization complexity
- Natural quantum dynamics transform temporal data into rich feature representations
- Suitable for noisy intermediate-scale quantum (NISQ) hardware → FTQ 2030
- Provides efficient processing of sequential and time-series data

QNIVERSE

IBM CONDOR - 1121
HERON - 433

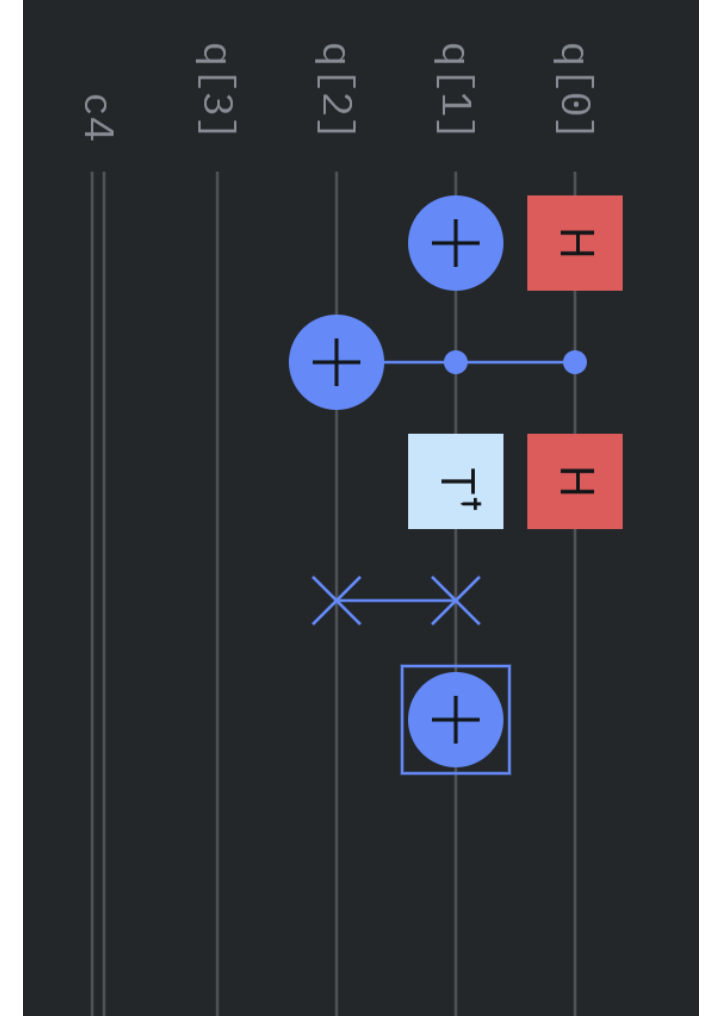
QRC Demo

- Encode sequential input data into a quantum reservoir using parameterized quantum states
- Allow the quantum system to evolve naturally without training its internal parameters
- Measure quantum states after each time step to extract feature vectors
- Train a classical regression or classification model on the measured outputs
- Evaluate prediction accuracy on unseen time-series data

[git@github.com:QuEraComputing/QRC-tutorials.git](https://github.com/QuEraComputing/QRC-tutorials.git)

Types of Quantum Computers

- **Gates Base QC**
 - Super Conducting (niobium+aluminium)
 - IBM (1121), Google, Rigetti
 - TIFR
 - Ion Trap (calcium or ytterbium)
 - IonQ
 - TCG-CREST
 - Photonic (photon)
 - Xanadu Borealis
 - IISC
 - Neutral Atom (rubidium 87)
 - QuEra, Atom Computing, CalTech (6000)
 - Topological
 - Microsoft (1 qubit)
- **Quantum Annealing**
 - D-Wave Quantum Annealer (4000 qubits)



QRC Applications

- Time-series forecasting in finance, weather prediction, and sensor analytics
- Speech recognition and natural language sequence processing
- Anomaly detection in cybersecurity and industrial monitoring
- Control systems and robotics requiring real-time decision making
- Scientific simulations involving complex dynamical systems

Backup Slides – Neutral Atom

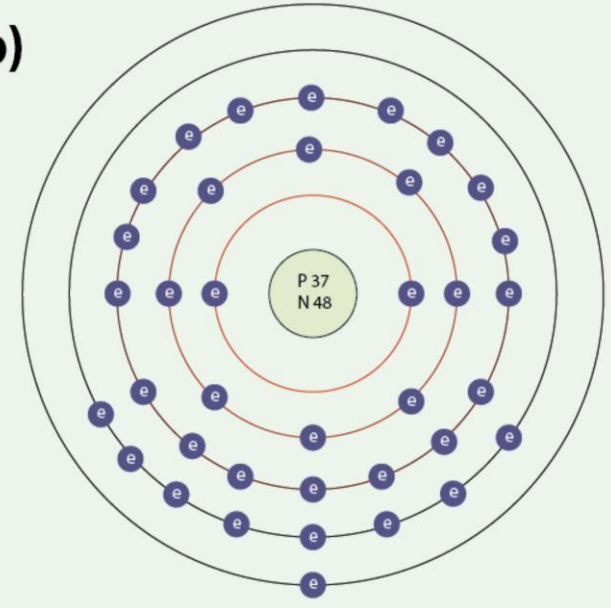
In neutral atom computers, the qubit is represented in the electronic state of the valence electron of Rubidium 87.

Arrays of individual atoms are held by laser tweezers, and quantum computations are executed by manipulating the electronic state of each atom using lasers and RF fields.

Entanglement can be generated using the [Rydberg state](#), which is a highly excited state that strongly interacts with adjacent atoms through a R^{-6} power law Van der Waals force.

Rubidium (Rb)

Energy levels: 5
Protons: 37
Neutrons: 48
Electrons: 37



Rydberg Atom

A **Rydberg atom** is an excited atom with one or more electrons that have a very high principal quantum number, n .

The higher the value of n , the farther the electron is from the nucleus, on average. Rydberg atoms have a number of peculiar properties including an exaggerated response to electric and magnetic fields, long decay periods and electron wavefunctions that approximate, under some conditions, classical orbits of electrons about the nuclei.

The core electrons shield the outer electron from the electric field of the nucleus such that, from a distance, the electric potential looks identical to that experienced by the electron in a hydrogen atom.

Planck constant
 $4.14 \times 10^{-15} \text{ eV/s}$

14-Apr :
World Quantum Day

Quantum-mechanically, a state with abnormally high n refers to an atom in which the valence electron(s) have been excited into a formerly unpopulated electron orbital with higher energy and lower binding energy. In hydrogen the binding energy is given by:

$$E_B = -\frac{R_y}{n^2},$$

where $R_y = 13.6 \text{ eV}$ is the Rydberg constant. The low binding energy at high values of n explains why Rydberg states are susceptible to ionization.

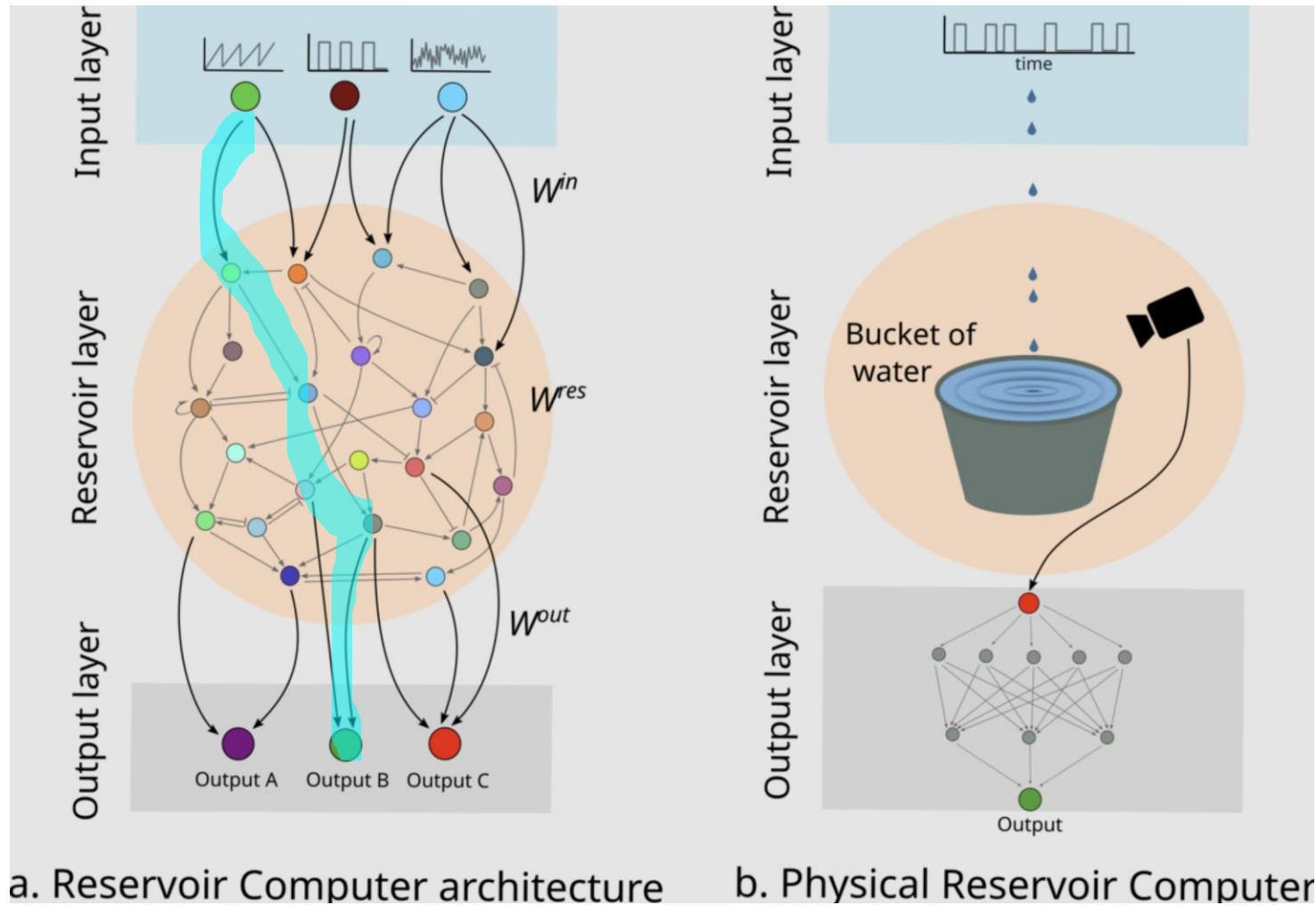
QuEra Bloqade Analog Mode

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{\mathcal{H}}(t) |\psi\rangle,$$

where $\mathcal{H}$ is a time-dependent "Rydberg atom" Hamiltonian.

$$\frac{\mathcal{H}(t)}{\hbar} = \sum_j \frac{\Omega_j(t)}{2} \left(e^{i\phi_j(t)} |g_j\rangle \langle r_j| + e^{-i\phi_j(t)} |r_j\rangle \langle g_j| \right) - \sum_j \Delta_j(t) \hat{n}_j + \sum_{j < k} V_{jk} \hat{n}_j \hat{n}_k,$$

Reservoir Computing



QuEra Bloqade Analog Mode

QRC extends classical RC by using **quantum systems** as reservoirs. Quantum reservoirs exploit **superposition, entanglement, and high-dimensional Hilbert spaces** to encode and process information in ways that classical systems cannot [arXiv.org](#). Inputs are injected into the quantum system, and measurements of the system's state serve as the readout. This allows QRC to capture both linear and nonlinear features of temporal data efficiently [arXiv.org+1](#). Key features of QRC include:

High-dimensional processing: Quantum states provide exponentially larger computational spaces than classical reservoirs.

Natural dynamics: Many-body quantum interactions generate complex, nonlinear dynamics suitable for computation [Phys.org+1](#).

Reduced training requirements: Some QRC schemes can achieve reliable predictions with fewer training data by exploiting quantum correlations [arXiv.org](#).

Edge-of-chaos operation: QRC performance is often maximized near the boundary between order and chaos in quantum many-body systems, similar to classical reservoirs [Phys.org](#).

Bloqade - Reservoir Computing

- [QuEraComputing/QRC-tutorials: A set of tutorials for quantum reservoir learning.](#)

Quantum Encoding

- Classical Information should be encoded into Quantum Information
- There are 4 major encoding systems
 - Basis Encoding
 - Angle Encoding
 - Amplitude Encoding
 - $P_x|x\rangle$
 - Hybrid Encoding
 - $P_x|x\rangle\langle x|$
 - State preparation
- Supervised Machine Learning using Quantum Computing by Maria Schuld

Books

- Quantum Machine Learning by Peter Wittek
- Quantum Machine Learning using Python by Shantanu Ganguly
- Quantum Machine Learning online by
- Online Learning documentation from
 - IBM
 - PennyLane
 - Quera